

Pritish Ranjan

B.Tech, Data Science & Engineering · IIT Mandi

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[linkedin](#) | [github](#) | [portfolio](#)

EDUCATION

Indian Institute of Technology, Mandi

August 2024 – June 2028

B.Tech, Data Science & Engineering — CGPA: 8.34 / 10.00

Himachal Pradesh, India

- **Relevant Coursework:** Programming and Data Structures, Probability and Statistics, Machine Learning, Design of Algorithms, Computer Organisation, Mathematical Foundations of Computer Science, Statistical Foundations for Data Science, Deep Learning, Matrix Computations for Data Science, Computing systems for data processing

PROJECTS

Finemonix – SME Financial Intelligence Platform

[GitHub](#)

- **Tech Stack:** Next.js, FastAPI, PostgreSQL, Neo4j, PyTorch, XGBoost, SHAP
- Built an AI-powered SME finance platform across 5+ modules for cash flow forecasting, credit scoring, and client risk monitoring.
- Implemented LSTM-based forecasting with **91% accuracy** and anomaly detection flagging **95%** of irregular transactions using XGBoost + SHAP explainability.
- Designed scalable architecture with FastAPI microservices, Neo4j graph analysis over 10K+ entity relationships, and real-time dashboards under **200ms** latency.

NLQ on Ego4D – Video Temporal Grounding

[GitHub](#)

- **Tech Stack:** Python, PyTorch, Transformers, BERT, CLIP, Ego4D
- Modified VSLNet and UniVTG architectures for natural language video temporal grounding, improving R@1 IoU=0.5 by **6.3%** over baseline.
- Implemented transformer-based multimodal fusion with bidirectional attention and multi-scale temporal reasoning, reducing grounding error by **18%**.
- Trained and evaluated pipelines on 3,670+ hours of egocentric footage from the Ego4D benchmark across 9 query categories.

RailSafe System – AI Railway Safety Platform

[GitHub](#)

- **Tech Stack:** React, Node.js, Python, YOLOv8, OpenCV, Socket.IO, Gemini API
- Developed a real-time railway safety system with vibration analytics and computer vision-based incident detection achieving **93% precision**.
- Integrated YOLOv8 segmentation across 6 risk zones for intrusion detection, cutting false positive rate by **35%** over threshold-based methods.
- Built live monitoring dashboards processing 30 FPS video streams with AI-generated incident summaries and **sub-2s** LLM-powered alert response.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: FastAPI, React.js, Next.js, PyTorch, Flask, Node.js, Tailwind CSS, Express.js

Domains & Methodologies: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Full-Stack Development, Object Detection, Time-Series Forecasting

Tools, Databases & Platforms: Git, GitHub, Redis, PostgreSQL, MongoDB, Neo4j, Docker, Linux, VSCode, Vercel

POSITIONS OF RESPONSIBILITY

Corporate & Relations Team Member, Xpecto'25 – IIT Mandi

November 2024 – March 2025

- Coordinated sponsorship outreach, brand collaborations, and execution of technical and entertainment events attended by 1000+ participants.

Core Team Lead, Music Club – IIT Mandi

September 2024 – Present

- Spearheaded operations, organized large-scale musical performances, and drove student engagement events for the campus community.

ACHIEVEMENTS

- Advanced to the penultimate evaluation round in the **Hack4Delhi** hackathon, securing a **Top 15** position in the domain among 900+ participating teams.
- Shortlisted among top competing teams in the **AI for Bharat** innovation challenge.